

# Virtual reality — the new media?

A S Rogers

---

*Multimedia computing is becoming increasingly prevalent in both the home and the office. Multimedia information is usually presented on the computer screen using 2-D menus and windows, but improving graphics capabilities are opening up new opportunities to present information within a 3-D virtual world. This paper describes the work of the Virtuosi project which is investigating the use of virtual reality to support groupworking. Applications include finding people, viewing information and interacting with colleagues in a virtual world. The 3-D approach is very powerful and could change completely the way we interact with people and information.*

---

## 1. Beyond multimedia

When multimedia systems started appearing there were many people who could not see what advantage it would be to be able to play sound from a PC, display high quality pictures on the screen or see a movie in a window. However, as illustrated throughout the other papers in this issue, multimedia is growing in importance and many multimedia applications are emerging. Three main areas for multimedia applications can be identified. One area is concerned with finding information and is exemplified by multimedia encyclopaedias, sales catalogues, image libraries, etc. Another area is the presentation of information in a structured way for training, sales promotions, etc. The third area, which is now beginning to make an impact on the market, is desktop conferencing which enables people to interact with each other using video as well as voice.

The use of multimedia, whether it includes sound, still image or moving pictures, can convey a very rich message to the user. However, in our home and office environments we not only make use of a variety of '2-D' media but also work in three dimensions. We spread things around so that the things we are currently using are close to us while other things we may need later are further away and less intrusive, but still reachable when required. In a multimedia system, all information is presented to us in 2-D windows, overlapping on a 2-D screen. This screen can quickly become very cluttered and it is often not as convenient to use as a set of books and papers on our desks. Similarly if we are in a room with a large number of people we will be aware of people in different parts of the room, but if we want to speak to someone in particular we move closer to

them to catch their eye when it is convenient. Desktop conferencing systems provide the opportunity to see as well as speak to other people and share information with them, but it is still very much a flat world with the picture of the other person fixed in a given place on a 2-D screen.

However, as computing power continues to grow, it is now becoming possible to present information, not just in a 2-D multimedia world, but also in a 3-D virtual reality world. This enables information to be presented in a much more natural way and opens up many opportunities to improve the ways that users can interact with information and with other people via their computers. This paper describes some of the ways in which a project called Virtuosi, led by BT, is developing virtual reality and 3-D visualisation techniques to support people in their day-to-day work and communications.

## 2. Virtuosi project background

Virtuosi [1] is a collaborative project involving a number of industrial and academic partners. The project is led by BT and the other industrial partners are GPT Limited with GMMT Hirst division (formerly Hirst Research Centre) as a sub-contractor, BICC plc and Division Ltd. The academic partners are the Universities of Nottingham, Lancaster and Manchester. Also associated with the project are Nottinghamshire County Council and Nottingham Trent University supporting operations in the fashion industry. The project is part of a government programme in Computer Supported Co-operative Work (CSCW) which is jointly

funded by the Department of Trade and Industry (DTI), the Engineering and Physical Sciences Research Council (EPSRC) and the industrial collaborators.

The sub-title for the project is 'support for virtual organisations'. In this context, the term virtual organisation refers to a group of people who, although formally part of a number of different companies or organisations within the same company have come to work together with a particular set of shared objectives. The formation of the virtual organisation might be explicit, for instance a project team within a company, or it may be more informal in the form of some sort of special interest group. Good communications are vital for a virtual organisation to achieve its goals. It might seem ideal for everyone to work together in the same office. However, this may not be practical or even desirable since relocation can bring its own problems and disruption. Some form of electronic support is required that can give all the benefits of sharing the same open plan office, but hopefully without some of the disadvantages. *Virtuosi* is investigating how such support could be provided in two main application areas. One is in an industrial manufacturing setting where the aim is to form the technical experts and the staff in a number of different factories into a much closer, collaborating community. The other is in the fashion and clothing industry in which designers, buyers, manufacturers and many other specialists are all involved in the complex and time-critical job of creating and selling a garment.

### 3. Virtual reality to support groupworking

**M**ultimedia is already being used to support group working. The most obvious application of this is desktop videoconferencing. This provides a voice and video connection between two users and also enables them to transfer files and share applications. However, full support of group working within a virtual organisation requires much more than just being able to set up point-to-point communications. It involves the creation of a complete environment that will enable users to interact intuitively and to feel as though they are sharing an office with the other members of the team. There are three main functions that cover most of these requirements. These are really extensions of three main types of multimedia application already identified — finding and viewing information, presenting and interpreting information and interacting with people.

#### 3.1 Finding and viewing information

##### The factory pilot

Many functions within a business environment are concerned with locating and retrieving information. As the amount of information grows this can prove to be a difficult task especially when the information is generated and shared by a large number of people. In the factory pilot a study is being made of the operations of a technical centre which provides support for a number of factories across the

world. An important resource generated by members of the technical team is the number of papers and reports that document the various studies and investigations in which they are involved. The operations of the technical centre have been studied by sociologists from Lancaster University using ethnography, a technique which concentrates on observing the interactions between people and their environment. The studies at the technical centre have shown the particular importance of documents in the work of the centre not only as a means of communication but also for structuring the work they do.

The documents produced by the centre are now being scanned and stored in a database to make them more accessible both to people inside the centre and to factories around the world. The initial implementation of the documentation service will provide a standard set of indexing facilities using keywords and text searching of abstracts, etc. However, studies are underway in *Virtuosi* to investigate other ways of finding and viewing the reports. One method being studied is called populated information terrains (PITs) (see Fig 1). This uses 3-D to represent different features or parameters of the documents. By choosing suitable parameters for the axes, documents of a similar type will be seen to be clustered together. Colour and shape can also be used to enable the user to see general characteristics of a document at a glance. PITs have another interesting feature in that they not only represent the information in the database in a 3-D manner, but they also provide indications about other people accessing the data. A normal database does not give any indication about other users of the system. Indeed the environment is deliberately designed so that each user is entirely independent and unaware of other users. However, it may be helpful for a user to know if anyone is accessing the same report or cluster of reports at the same time since they may be able to help with a particular problem. Similarly a history of previous accesses may indicate other users with similar interests.

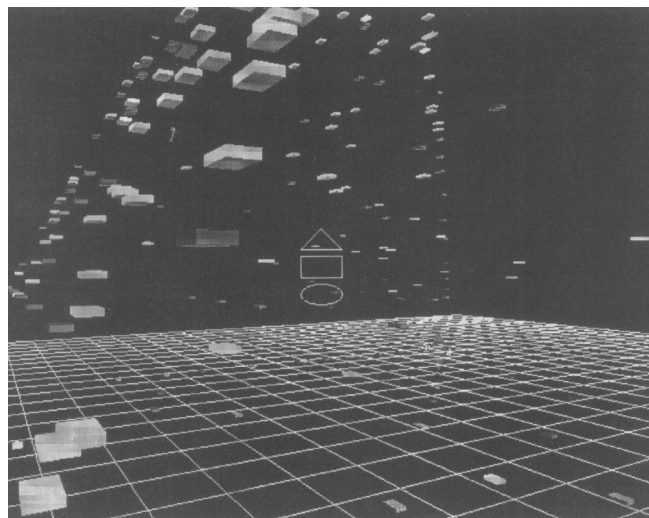


Fig 1 A populated information terrain.

This link between information and people is important. Moreover, in many cases, documents are used not so much as a source of information in their own right as links to people who can help. Indeed, finding people within a real or virtual organisation is as important if not more so than finding information. It is also a very dynamic issue since the availability of people is continually changing and a user may wish to communicate with different people depending on who is available.

If a group of people are working together in the same office, it is very easy to see who is available. People coming into the office may approach a person they know can help them or someone close by if the first person is not there. The factory pilot is looking at ways to provide this functionality in a virtual environment. An early prototype that has been developed is a visualisation of one of the offices within BICC. It is not a fully accurate representation but it gives a good impression of the layout of the office. Each desk is labelled with the name of the person who usually sits there. This world can be 'visited' by a user who can move about freely in the virtual space. It is a shared environment, so more than one user can be present in the virtual space and they are represented by very angular shapes which can at least show where they are and in what direction they are looking (see Fig 2).

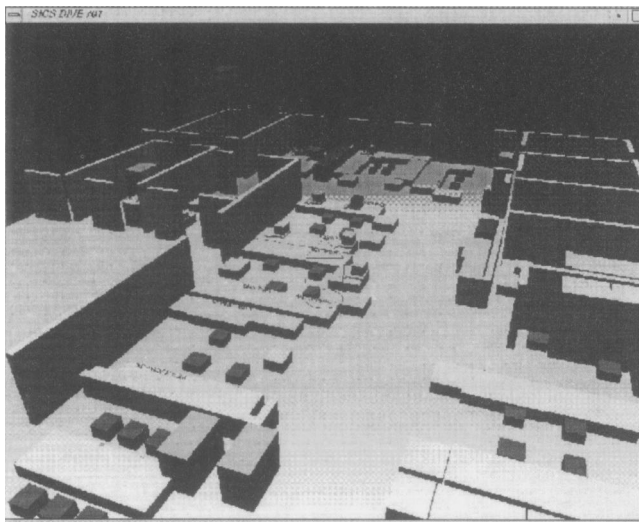


Fig 2 Office flythrough.

#### A virtual project office

The next stage in the development is to provide more information about the people within this world. One aspect is to be able to indicate whether people represented in the world are really available or not. A 'Rich Finger' server, or 'Activity Monitor' server is being developed to achieve this. The design draws inspiration from the 'Finger' service in UNIX systems which allows one user to 'finger' other users to see if they are logged on, how active they have been recently and the state of their mailbox. The 'Rich Finger'

will provide similar facilities for users on a range of different types of terminal. In addition to computer-based information, the system will also extract information from the private telephone system to indicate if they are on the phone now, or have been recently. The best way to present this information, to a user browsing a virtual office, is still a topic for research within the project, but could well include the use of a range of colours to indicate the degree of availability with pop-up windows giving fuller details, if required.

An important aspect of this approach to browsing is that the office does not need to represent a real office at all. The visualisation could just as easily be used to represent a group of people who are working together but are actually distributed in many different places. One demonstration currently being developed is the Virtuosi project office. This will provide a 'Virtual open-plan office' with groups of desks representing the various partners in the project. Anyone working on the project will be able to enter the space and see who else in the project is available. The aim is to provide a much better feeling of working together than can be achieved normally with members of a project team located in many different places.

#### The fashion pilot

A different aspect of browsing for information is illustrated in the fashion pilot. In the fashion industry, as in many industries concerned with meeting consumers' demands, an important part of the design process is sourcing the various components of a product. In the clothing industry this may include yarns, textiles, accessories, etc. Depending on the position in the production chain, designers may need to assess the latest trends to incorporate in their designs, buyers may need to find suitable designers and manufacturers for their goods and manufacturers may be looking for sub-contractors. Associated with the fashion pilot, Nottingham Trent University is developing a database of fashion information (see Fig 3). The initial database is being constructed using a standard database package. However, studies are under way, to see how such an information brokering service could be extended to support the virtual organisations that are effectively set up every time a garment is designed, manufactured and sold. The development of such a service raises issues such as ownership of information, billing, efficient searching and retrieval and many others.

As with other aspects of the project, 3-D may have a role to play. One possibility being investigated is representing the user interface to the information as a 3-D fashion trade show rather than a set of windows and menus (see Fig 4). Not only would this make searching for information more acceptable for non-technical users, it would also enable users to see other people browsing the

database, opening up the opportunity for them to discuss what they are seeing and to share discoveries.

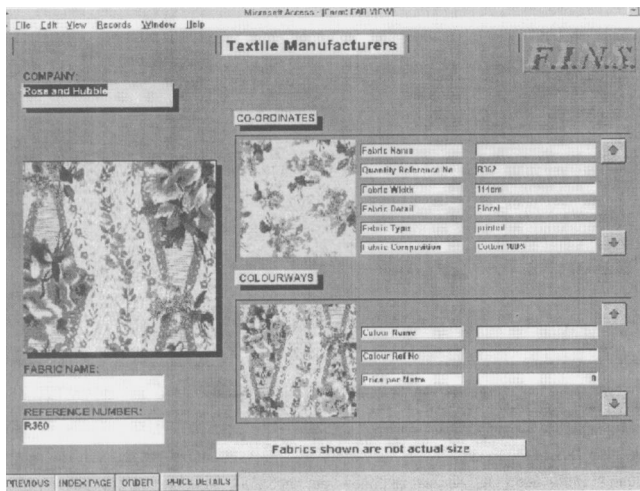


Fig 3 A page from the fashion database.

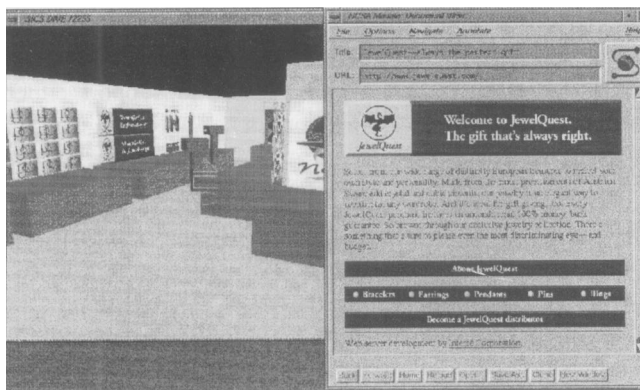


Fig 4 A virtual fashion show.

### 3.2 Viewing and presenting information

3-D and virtual reality also have a useful role to play in viewing and presenting information. The previous section has already shown that the 3-D paradigm is a very useful way of helping people to search for information. One reason for this is that humans are able to think and remember in spatial terms. This is also useful when presenting complex information because facts and trends that would have been hidden in a table of figures or even in a set of graphs can suddenly become very obvious when presented using a spatial metaphor.

The ideal way for collaborators to discuss a 3-D visualisation of some information would be in a 3-D world.

This has been investigated in Virtuosi. One example was a visualisation of a database structure which became very clear when shown as a 3-D model (see Fig 5). The model was generated in a virtual environment and two users were able to move about the structure and discuss its details.

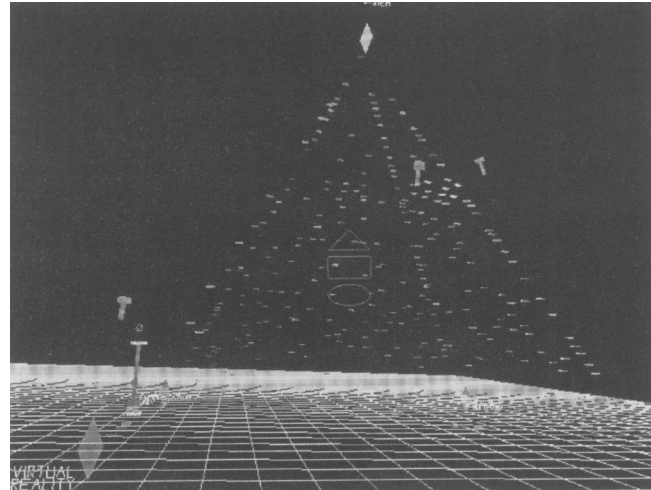


Fig 5 Database visualisation.

One area very relevant to 3-D design is the fashion industry. The product of the industry is clearly a 3-D object. However, a garment is a very complex object. It is constructed from a number of 2-D shapes (the pattern pieces), but it does not have a shape purely on its own since it takes its shape from the person wearing it. Modelling a garment is thus a very complex problem involving the physical and mechanical properties of the cloth. Work is under way at Nottingham Trent University to model cloth and garments. The ultimate aim is to be able to create a virtual catwalk (see Fig 6) with the mannequin and the garment moving in an entirely natural way. The computing power needed to achieve this will certainly be very great although not totally inconceivable, given current trends in computer development. More immediately, work is concentrating on achieving a realistic hang and drape on a stationary model. Tools are being developed to enable modifications to be made to a 3-D garment and then to recalculate the details of the garment. The 3-D garment will be available for viewing in a virtual world and it will be possible to share this view with a number of other people in the virtual world, all able to vary their own view point and to see the others who are sharing the world. To simplify the control of the mannequin, a voice recognition system is being developed so that users will be able to give spoken commands to direct the mannequin.

This technology raises some interesting possibilities in the field of home shopping. Using these techniques it will be possible to construct the mannequin with the exact

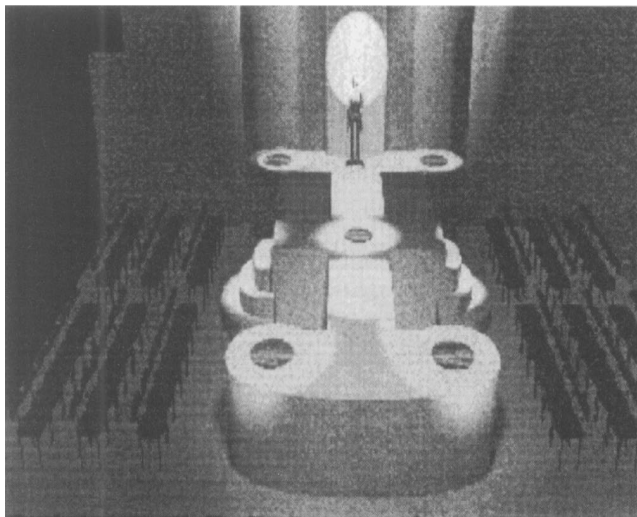


Fig 6 A populated office.

dimensions of an individual customer. The customer would be able to see remotely if the garment fits and, if necessary, the garment could be altered to fit exactly. The system could then produce bespoke garments by using the 3-D representation to produce the 2-D patterns that would be passed directly to the manufacturing unit to make the garment to the customer's exact specifications.

### 3.3 Interacting with people

In describing the use of 3-D for finding people and information and in presenting and viewing information, the fact that other users can inhabit the same virtual space has been mentioned a number of times. This opens up many new ways in which people may interact with each other remotely. One of the most important possibilities that affords itself is that people can use space to control their remote interactions in much the same way as they do in face-to-face meetings. Making a telephone call may be compared with knocking (loudly) on someone's door and entering when invited. However, many of our interactions are much more complex than this. On entering a room full of people we may choose who to speak to depending on who is already in conversation and with whom. We also find it useful to use the direction of a sound to help us monitor who is speaking and to listen to a particular conversation out of a number which may be going on.

To achieve this in a virtual world, a number of features must be provided. The first is to be able to represent other people in the virtual world so that they can be recognised (see Fig 7). This does not necessarily mean that there needs to be a perfect likeness. Experiments within *Virtuosi* and other projects show that even a very basic block representation of a person can still give useful clues about other people in the environment and, with some visual indication when they are speaking, can enable useful conversation to take place.

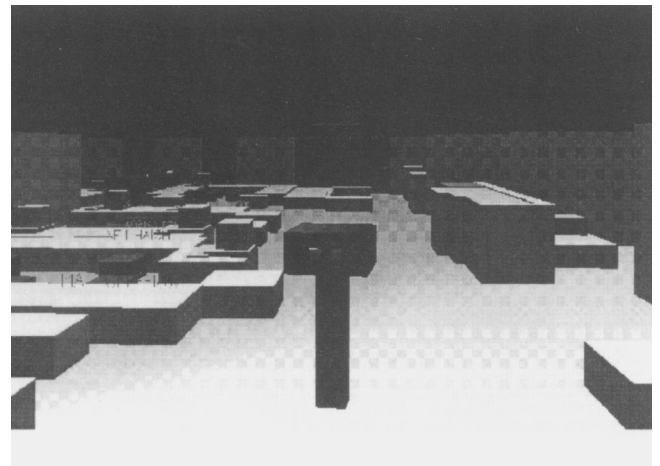


Fig 7 A populated office.

Another obvious feature that is needed for interaction between people in the virtual world is the ability to hear each other. In the real world we do not hear the people in a large room equally loudly. Those close by are clearer than those further away and indeed some people will be seen but not heard. This has to be mimicked in the virtual world by controlling audio connections by the relative positions of users in the environment. Initial experiments have required audio connections to be set up as users enter a specific virtual world. However, an audio service is now being developed within *Virtuosi* which will automatically set up the connections when two or more people approach within a certain distance of each other. The architecture of the system is being designed to enable it to support either packet audio over the data network or the use of an overlay of telephone connections.

An important aspect of the system design is the decision whether to use a fully distributed approach or to use a central audio server. The distributed approach with multicasting of audio packets is the most flexible approach. However, in environments where multicast is not possible, a distributed system with a large number of users in a single conversation would require a large number ( $\sim n^2$ ) connections. In these cases, a centralised server would be more practical.

Techniques are available which would enable the volume of the sound not only to be controlled but also its position in the virtual space to be fixed. This would add considerably to the complexity of the audio server, but would greatly enhance the feeling of really 'being there'. *Virtuosi* will be investigating spatialised sound and the benefits it can bring to enhance user interaction.

## 4. Conclusions

The *Virtuosi* project is tackling a range of issues concerned with providing support to groups of people working together. The use of virtual reality opens up a wide

range of new ways of enabling people to interact with information and with other users. Using virtual reality it is possible to mimic the way in which we interact when we are in the same room much more closely than can be achieved with existing means of communication, such as the telephone or even videophone. A very important issue is integrating a range of different tools so that co-operation and sharing can be achieved as seamlessly as possible.

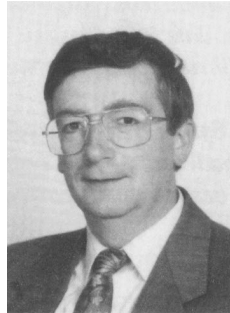
Although virtual reality can be used to mimic the real world, one of its major advantages is the increased control that users can have over their own environment. For instance, physical space is no longer a constraint and areas within the virtual environment can be expanded or contracted at will. It is also possible to create features such as 'secrecy zones' so that conversations that would otherwise be overheard, can be held in secret. An important area of research is to determine which facilities will be most useful in the virtual environment.

The Virtuosi project has been concentrating on presenting the virtual environment to users by means of conventional workstation displays. As the technology improves, there will also be opportunities to use fully immersive techniques for co-operative working. This will improve the realism of the interactions and increase the

feeling of really 'being there'. The thought of putting on a VR headset and interacting with our colleagues, represented as angular blocks with eyes, seems rather strange at present, but the principles being investigated in Virtuosi may well lead to the new generation of computing and communications systems which will take the next step beyond multimedia.

### References

- 1 Rogers A S: 'Virtuosi — support for groupworking', *BT Technol J*, **12**, No 3, pp 81—89 (July 1994).



Alastair Rogers joined BT as a sponsored student in 1969. After gaining a BSc(Eng) at Imperial College, London he commenced working on both analogue circuit design and system development for electronic relay set replacements. He took an MSc in Telecommunications Systems at Essex in 1979 and then worked on distributed processing gaining a PhD at Imperial College in 1987. He has led groups working on telecommunications architectures and distributed systems. He currently leads a technical group within the Centre for Human Communication at BT Laboratories and is the project manager for

Virtuosi, a £2.5m collaborative project using virtual reality technology to support groupworking.

He is a member of the IEE and a Chartered Engineer.